

EXER. A

G

(2)

s_1	←	-1	-9
s_1	←	-1	-8
s_1	←	-1	-7
s_1	↓	-1	-6
s_4	→	-1	-5
s_5	↓	-1	-4
s_8	←	-1	-3
s_7	↓	-1	-2
s_7	→	-1	-1
s_8	←	-1	0
s_7	↓	-1	1
s_7	→	-1	2
s_8	←	-1	3
s_7	↑	-1	4
s_4	↓	-1	5
s_7	↓	-1	6
s_7	→	-1	7
s_8	↓	-1	8
s_8	↓	-1	9
s_8	→	10	10

FIRST VISIT

$$V_{\pi}^{(FV)} =$$

-9	INIT. ESTIMATE 0	INIT. EST. -10
-5	-4	INITIAL ESTIM 0
-2	-3	INIT. EST. 10

EVERY VISIT

-9, -8, -7, -6		
-5, 5	-4	
-2, -1, 1, 2, 4, 6, 7	-3, 0, 3, 8, 9, 10	

$$V_{\pi}^{(EV)} =$$

-7.5	INIT. EST. 0	INIT. EST. -10
0	-4	INIT. EST. 0
2.43	4.5	INIT. EST. 10

④

$$V_{\pi}^{(EV)} =$$

-9	0	-10
-5	-4	0
-2	-3	10

$\Rightarrow$

π'

=

$\rightarrow$	$\uparrow$	10
$\downarrow$	$\uparrow \rightarrow$	$\downarrow$
$\leftarrow \downarrow$	$\rightarrow$	10

NOT OPTIMAL

SINCE P IS KNOWN, V_{π} IS ENOUGH TO DERIVE A NEW POLICY.

$$V_{\pi}^{(FV)} =$$

-7.5	0	-10
0	-4	0
2.43	4.5	10

$\Rightarrow$

π' :

NOT OPTIMAL

$\downarrow \rightarrow$	$\uparrow$	10
$\downarrow$	$\downarrow$	$\downarrow$
$\rightarrow$	$\rightarrow$	10

⑥

$$S_1, \rightarrow, -1 \quad G$$

$$-11 = G_1$$

$$S_2, \rightarrow, -10 \quad -10 = G_2$$

$$W_1 = \frac{\pi(\rightarrow | S_2)}{Q(\rightarrow | S_2)} = \frac{1}{6} \cdot 4 = \frac{2}{3}$$

$$W_2 = \frac{\pi(\rightarrow | S_2)}{Q(\rightarrow | S_2)} \cdot \frac{\pi(\rightarrow | S_1)}{Q(\rightarrow | S_1)} = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

$$Q(S_2, \rightarrow) = W_1 \cdot G_2 = -\frac{20}{3}$$

$$Q(S_1, \rightarrow) = W_2 \cdot G_1 = -\frac{44}{9}$$